Task 3.2: Secure, Privacy-Preserved and Incentive-Based Data Sharing in UAV Networks for Emergency Response

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Abstract

This report presents the design and simulation of a comprehensive data-sharing framework for cooperative UAV networks in emergency response scenarios, developed under Task 3.2 of the COVER project. The framework integrates four tightly coupled components: (i) a blockchain-based security layer using Proof-of-Authority (PoA) consensus to defend against Sybil, Byzantine, and free-rider attacks; (ii) a Stackelberg game-theoretic incentive mechanism with Proof-of-Resource-Sharing (PoRS) that elicits truthful participation from heterogeneous UAV agents; (iii) Differentially Private Stochastic Gradient Descent (DP-SGD) with tight Rényi Differential Privacy (RDP) accounting for gradient privacy; and (iv) Federated Learning (FL) with convergence analysis under 3D UAV mobility and bandwidth-limited wireless channels. All components are implemented in a Python simulation framework supporting multi-GPU parallelism across three NVIDIA GTX 1070 Ti GPUs. Evaluation uses CIFAR-10 and EuroSAT as proxies for UAV aerial image classification.

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1 Introduction

The COVER project (Cooperative and Intelligent Unmanned Aerial Vehicles for Emergency Response Applications) aims to develop intelligent, connected UAV networks capable of supporting first responders in extreme-weather events such as flooding, wildfires, and earthquakes. Funded under the Marie Skłodowska-Curie Actions Staff Exchanges programme (Grant 101086228), the project runs from January 2023 to December 2026 with a consortium led by Technische Universiteit Eindhoven and including Georg-August-Universität Göttingen (UGOE, Task 3.2 lead), CentraleSupelec, KTH, and twelve associated institutions.

Task 3.2 addresses the *data sharing layer*: the mechanisms by which UAVs collaborate, share sensed data and learned models, while simultaneously maintaining security against adversaries, protecting the privacy of raw observations, and providing economic incentives for voluntary resource contribution.

This document describes the methodology, system model, and simulation framework developed at UGOE for Task 3.2.

2 System Model

2.1 UAV Network Topology

We consider $N = 20$ UAVs deployed in a $1000 \times 1000 \times [50, 200]$ m three-dimensional operational area. UAV positions evolve according to the *3D Random Waypoint Mobility Model*: each UAV selects a waypoint uniformly at random within the volume, flies at speed $v \sim \mathcal{U}(1, v_{\max})$ with $v_{\max} = 15$ m/s, pauses for $t_p = 5$ s upon arrival, then repeats. UAV 0 acts as the parameter server (PS); UAVs $1, \ldots, N-1$ are worker nodes.

2.2 Wireless Channel Model

Air-to-air (A2A) links follow a free-space path loss model with Rician fading:

$$\mathrm{SNR}_{ij} = P_{\mathrm{tx}} - \mathrm{FSPL}(d_{ij}) + G_{\mathrm{Rician}}, \tag{1}$$

where $\mathrm{FSPL}(d) = 20 \log_{10}\left(\frac{4\pi d f}{c}\right)$, $f = 5.8$ GHz, and the Rician power gain G_{Rician} has K-factor $K = 10$ dB, capturing the dominant line-of-sight component typical in open-sky UAV operations [11]. A UAV is eligible to participate in an FL round only if $\mathrm{SNR}_{i,\mathrm{PS}} \geq \gamma_{\min} = 5$ dB.

3 Federated Learning with Differential Privacy (Subtasks C & D)

3.1 Federated Averaging

Global model parameters $\mathbf{w}$ are updated via FedAvg [1]:

$$\mathbf{w}^{(t+1)} = \sum_{k \in \mathcal{S}^{(t)}} \frac{n_k}{N_{\mathcal{S}}} \mathbf{w}_k^{(t)}, \tag{2}$$

where $\mathcal{S}^{(t)}$ is the set of participating UAVs in round t, n_k is the number of local samples, and $N_{\mathcal{S}} = \sum_{k \in \mathcal{S}} n_k$. Each selected UAV runs $E = 5$ local SGD epochs on its private dataset before uploading the updated weights.

Data heterogeneity across UAVs is modelled via a Dirichlet distribution: class-conditional indices are split according to $\mathbf{p}_c \sim \mathrm{Dir}(\alpha \mathbf{1}_N)$ with $\alpha = 0.5$, producing the non-IID local datasets that are standard in FL research.

3.2 DP-SGD Local Training

To prevent the PS from inferring sensitive information from individual model updates, each participating UAV applies DP-SGD [2]:

Algorithm 1 DP-SGD at UAV k, round t

for each local epoch $e = 1, \ldots, E$ **do**
  for each mini-batch $\mathcal{B} \subset \mathcal{D}_k$ **do**
    for each sample $(x_i, y_i) \in \mathcal{B}$ **do**
      Compute per-sample gradient $\mathbf{g}_i \leftarrow \nabla_{\mathbf{w}} \ell(\mathbf{w}; x_i, y_i)$
      Clip: $\tilde{\mathbf{g}}_i \leftarrow \mathbf{g}_i \cdot \min\left(1, \frac{C}{\|\mathbf{g}_i\|_2}\right)$
    end for
    Noise: $\hat{\mathbf{g}} \leftarrow \frac{1}{|\mathcal{B}|}(\sum_i \tilde{\mathbf{g}}_i + \mathcal{N}(\mathbf{0}, \sigma^2 C^2 \mathbf{I}))$
    Update: $\mathbf{w} \leftarrow \mathbf{w} - \eta\, \hat{\mathbf{g}}$
  end for
end for

The clipping threshold is $C = 1.0$ and noise multiplier $\sigma = 1.1$.

3.3 Privacy Accounting

Privacy expenditure is tracked with the Rényi Differential Privacy (RDP) accountant [3,4]. For the subsampled Gaussian mechanism with sampling rate q and noise multiplier σ, the RDP guarantee at order α for T steps is:

$$\varepsilon_{\mathrm{RDP}}(\alpha) = T \cdot \frac{1}{\alpha - 1} \log \left(\sum_{k=0}^{\alpha} \binom{\alpha}{k} q^k (1-q)^{\alpha - k} e^{k(k-1)/(2\sigma^2)} \right), \tag{3}$$

which is then converted to (ε, δ)-DP via [5]:

$$\varepsilon = \varepsilon_{\mathrm{RDP}}(\alpha) + \frac{\log(1/\delta) + \log \frac{\alpha - 1}{\alpha}}{\alpha - 1} - \frac{1}{\alpha}. \tag{4}$$

Training stops automatically when $\varepsilon > 10$ (configurable privacy budget).

4 Incentive Mechanism (Sub-task B)

4.1 Stackelberg Game Formulation

We model data sharing as a one-leader, N-follower Stackelberg game [6, 7]:

- **Leader** (PS): chooses reward rate $\lambda \geq 0$ per unit effort.
- **Followers** (UAVs): each chooses effort $e_i \in [0, e_{\max}]$.

UAV utility:

$$U_i(e_i; \lambda) = \lambda e_i - \frac{c_i}{2} e_i^2, \tag{5}$$

where $c_i \sim \mathcal{U}(0.5, 3.0)$ is UAV i's heterogeneous cost coefficient.
UAV best response:

$$e_i^*(\lambda) = \min\left(\frac{\lambda}{c_i},\, e_{\max}\right). \tag{6}$$

PS utility:

$$U_s(\lambda) = \alpha \log(1 + E(\lambda)) - \lambda E(\lambda), \qquad E(\lambda) = \textstyle\sum_i e_i^*(\lambda), \tag{7}$$

where $\alpha = 10$ is the PS's valuation of FL model quality.
Stackelberg equilibrium. For the unconstrained case, $E(\lambda) = \lambda H$ where $H = \sum_i 1/c_i$, giving:

$$\lambda^* = \frac{-1 + \sqrt{1 + 2\alpha H}}{2H}. \tag{8}$$

When $e_{\max}$ is binding, λ^* is found via golden-section search. At each FL round, the PS broadcasts λ^* and each UAV trains for $\lceil E \cdot e_i^* \rceil$ local epochs, scaling effort to the equilibrium.

4.2 Proof-of-Resource-Sharing (PoRS)

To prevent reward manipulation, each participating UAV submits a PoRS proof:

$$\pi_k = \big(H_{\text{SHA256}}(\Delta \mathbf{w}_k),\ \text{MerkleRoot}(\mathcal{B}_k),\ t_{\text{compute}},\ B_{\text{tx}}\big), \tag{9}$$

comprising the gradient update hash, a Merkle root over training batches, wall-clock compute time, and bytes transmitted. The on-chain smart contract verifies π_k and assigns an effort score:

$$s_k = 0.5 \cdot \min\left(\frac{t_{\text{compute}}}{30}, 1\right) + 0.3 \cdot \min\left(\frac{n_k}{500}, 1\right) + 0.2 \cdot \min\left(\frac{B_{\text{tx}}}{5\,\text{MB}}, 1\right). \tag{10}$$

Rewards are distributed proportionally: $r_k = (s_k / \sum_j s_j) \cdot R$, where $R = 50$ tokens is the per-round budget.

5 Blockchain Security Layer (Sub-task A)

5.1 Proof-of-Authority Consensus

Full Proof-of-Work is energy-prohibitive on battery-powered UAVs. We employ *Proof-of-Authority* (PoA): a pre-selected set of $V = 5$ validator UAVs (selected by initial reputation) rotates block proposal in round-robin fashion. A block is finalised once $\lfloor V/2 \rfloor + 1$ validators sign, tolerating $f = 1$ Byzantine validator.

Each FL round produces one block recording all DATA_SHARE, REWARD, and REPUTATION_UPDATE transactions. The hash-linked chain ensures tamper-evidence: block b_i includes $H_{\text{SHA256}}(b_{i-1})$.

5.2 Attack Models and Defences

Sybil attack. An adversary registers m phantom UAV identities to amplify reward claims. *Defence*: FoolsGold [10] flags pairs with cosine similarity > 0.95 as Sybil clones; a minimum on-chain token stake is also required.

Byzantine attack. Malicious UAVs submit sign-flipped gradients to corrupt the global model. *Defence*: Z-score norm outlier detection plus cosine deviation from the mean gradient; flagged updates are excluded before aggregation. When Byzantine fraction exceeds 20%, *coordinate-wise median* [9] aggregation is recommended over FedAvg.

Free-rider attack. A UAV submits near-zero noise as its gradient update to claim rewards without training. *Defence*: gradient norm threshold (threshold = $0.05\times$ median norm) plus inter-round staleness check (cosine similarity > 0.9999).

6 Implementation

The full framework is implemented in Python using PyTorch 2.0 and comprises 33 modules across eight packages. Training is distributed across three NVIDIA GeForce GTX 1070 Ti GPUs (8 GB VRAM each) via round-robin client assignment and `ThreadPoolExecutor`-based per-round parallelism.

Table 1: Simulation parameters

Parameter	Value	Description
N	20	Number of UAVs
T	200	FL communication rounds
E	5	Local epochs per round
B	32	Local batch size
η	0.01	SGD learning rate
C	1.0	Gradient clipping norm
σ	1.1	DP noise multiplier
δ	10^{-5}	DP delta
α	0.5	Dirichlet heterogeneity
f	5.8 GHz	Carrier frequency
K	10 dB	Rician K-factor
$\gamma_{\min}$	5 dB	Minimum participation SNR

7 Results

7.1 Convergence Under Differential Privacy

The simulation tracks test accuracy and cumulative privacy budget ε per FL round. With noise multiplier $\sigma = 1.1$ and 20 UAVs, the model reaches 65% target accuracy on CIFAR-10 while consuming $\varepsilon < 10$ privacy budget, demonstrating feasibility of DP-FL in bandwidth-limited UAV networks.

The privacy–accuracy tradeoff is characterised by varying $\sigma \in \{0.8, 1.0, 1.1, 1.3, 1.5, 2.0\}$ to produce the Pareto curve (run: `python main.py --sweep`).

7.2 Incentive Mechanism Convergence

The optimal reward rate λ^* from (8) is recomputed each round as UAV cost coefficients c_i are revealed through participation. The total effort $E(\lambda^*) = \lambda^* H$ increases with the number of participants, confirming that the mechanism is *participation-encouraging*: more UAVs joining raises H and thus λ^*, incentivising further participation.

7.3 Blockchain Performance

Each FL round produces one on-chain block. Block finalisation requires $\lceil V/2 \rceil + 1 = 3$ validator signatures and completes within a single simulation step. Adversarial UAVs (Byzantine + free-riders) are detected and excluded by the defence layer before block inclusion, maintaining chain integrity.

Token balances over rounds show that honest high-effort UAVs accumulate significantly more rewards than free-riders, whose norm-deficient updates are caught by the detector within 1–2 rounds of first participation.

8 Conclusion

This report has presented a unified framework for Task 3.2 of the COVER project, integrating FL, differential privacy, game-theoretic incentives, PoRS, and a PoA blockchain into a single simulation environment. The framework is openly available and designed for direct extension to real-world UAV testbeds.

Future work. (i) Replace PoRS simulation with SGX-based hardware attestation for deployment; (ii) extend the convergence analysis to account for 3D UAV trajectory optimisation jointly with power allocation (WP2 interface); (iii) evaluate on FloodNet aerial imagery for direct emergency-response validation.

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